

Andrew Liu

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EDUCATION

University of California, Santa Barbara
College of Engineering - B.S. Electrical Engineering
Expected Graduation: June 2028
Cumulative GPA: 3.98/4.00

Relevant Coursework:

Digital & Analog Circuits, Circuits & Physics Labs, Digital Design Principles, E&M, Photonics, Modern Physics
Currently Taking: Signals & Systems, Solid State Electronics, VLSI, Fields & Waves
Linear Algebra, Differential Equations, Vector Calculus
Data Structures & Algorithms, Object-Oriented Programming, Probability & Statistics

WORK EXPERIENCE

Electrical Engineering Intern | Hoplynk June 2026 - Present

- Designed, debugged, and integrated Power Distribution & Battery Monitoring Printed Circuit Boards
- Designed and integrated power distribution systems using COTS electronics under time constraints
- Created write-ups and documents to explain my design choices, allowing for smooth design reviews
- Assembled and tested various electronics using a DMM and running test software (e.g., radios)

Avionics/Electrical Team Lead | Gaucho Rocket Project January 2026 - Present

- Design data acquisition systems and boards to log and stream rocket telemetry during flight & tests
- Verify electronic components and sensor functionality with lab test equipment (DMM, Oscilloscope)
- Document test results and create protocols so the team has reference data and test instructions

PROJECTS

Hoplynk Power Distribution & Battery Monitoring PCB (PCB Design) June 2026 - September 2026

- Designed a power distribution & battery charging board with DC inputs (**12V-32V**) + smart charger
- Used MOSFET transistors at IC pins to ensure pack-level isolation during battery discharge & charge
- Optimized board size while also derating all components to below 50% of max voltage for reliability
- Ensured EMI from high-power switching areas did not affect analog readings and digital ICs
- Implemented a digital switch to allow cut-off of a heavy load, increasing system runtime by **~86%**
- Wrote and flashed STM32 firmware and Python scripts to send battery data and commands via USB

Gaucho Rocket Project Sensor Logger Printed Circuit Board (PCB Design) April 2026 - May 2026

- Designed a 4-layer PCB in KiCad to log temperature, pressure, & acceleration data during rocket flight
- Selected integrated circuits for size, power, sensor compatibility, and avionics requirements
- Validated and tested key components and sensors (e.g., thermocouple amplifier) on breakout boards

Boost Converter Printed Circuit Board (KiCad, Soldering, Power Electronics) March 2026 - April 2026

- Designed and prototyped a boost converter circuit, achieving an efficiency of **~66%** (5V to 12V)
- Implemented pulse width modulation control with closed-loop voltage feedback using Arduino/ATtiny
- Designed PCB layout, taking into account trace width calculations and component placement

SKILLS

Technical Skills: Analog and Digital Circuit Analysis, Circuit Prototyping and Debugging, KiCad, PCB Design & Rework, Power Electronics, LTspice, C++, Python, Oscilloscope, Multimeter, Function Generator